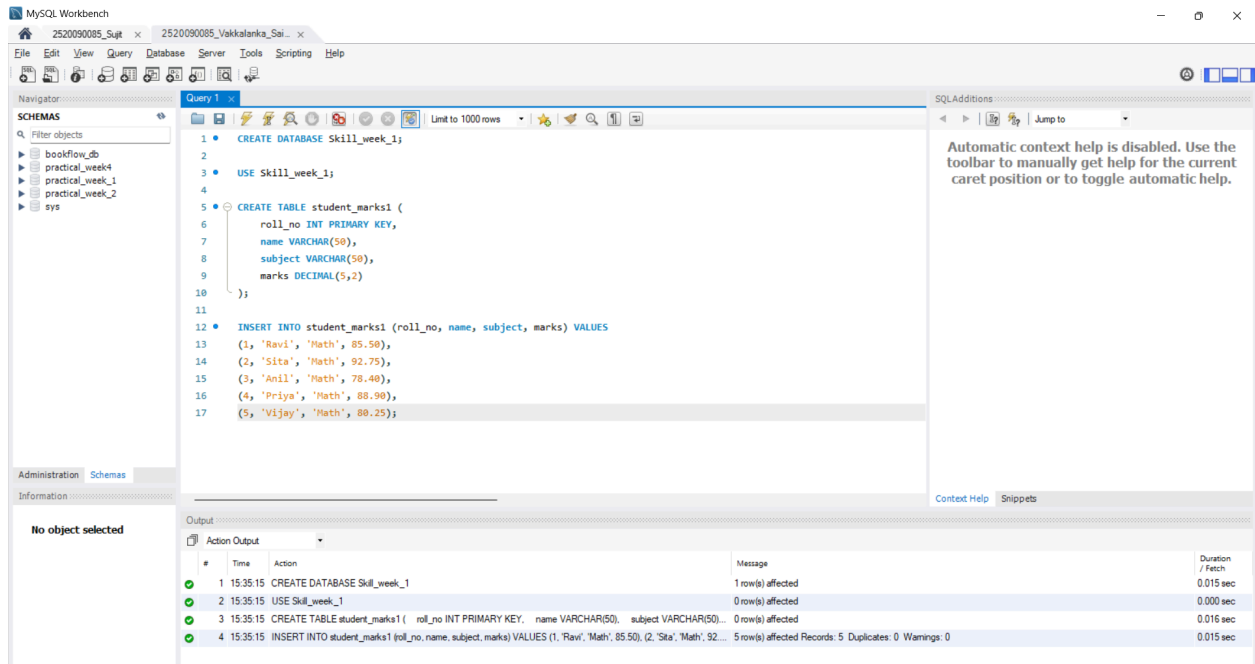


DBSE&DBD Skill Week-1

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1. Creating Database, Table and Inserting Data Into Table



The screenshot shows the MySQL Workbench interface with the following SQL queries in the Query Editor:

```
1 CREATE DATABASE Skill_week_1;
2
3 USE Skill_week_1;
4
5 CREATE TABLE student_marks1 (
6     roll_no INT PRIMARY KEY,
7     name VARCHAR(50),
8     subject VARCHAR(50),
9     marks DECIMAL(5,2)
10 );
11
12 INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
13 (1, 'Ravi', 'Math', 85.50),
14 (2, 'Sita', 'Math', 92.75),
15 (3, 'Anil', 'Math', 78.40),
16 (4, 'Priya', 'Math', 88.90),
17 (5, 'Vijay', 'Math', 80.25);
```

The Output window at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	15:35:15	CREATE DATABASE Skill_week_1	1 row(s) affected	0.015 sec
2	15:35:15	USE Skill_week_1	0 row(s) affected	0.000 sec
3	15:35:15	CREATE TABLE student_marks1 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50)...	0 row(s) affected	0.016 sec
4	15:35:15	INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec

2. Basic Selections and Aggregation Functions

MySQL Workbench

2520090085_Sujt x 2520090085_Vakkalanka_Sai...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- sys

Query 1

```

1 • SELECT * FROM student_marks1;
2 • SELECT COUNT(*) AS total_students FROM student_marks1;
3 • SELECT SUM(marks) FROM student_marks1;
4 • SELECT AVG(marks) FROM student_marks1;
5 • SELECT MAX(marks) FROM student_marks1;
6 • SELECT MIN(marks) FROM student_marks1;

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Filter Rows: Exports: Wrap Cell Contents

MDN(marks)

70.40

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	15:35:15	INSERT INTO student_marks1 (rol_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Mat...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
5	15:35:55	SELECT * FROM student_marks1 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
6	15:35:55	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	15:35:55	SELECT SUM(marks) FROM student_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
8	15:35:55	SELECT AVG(marks) FROM student_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
9	15:35:55	SELECT MAX(marks) FROM student_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
10	15:35:55	SELECT MIN(marks) FROM student_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

student_marks1 1 Result 2 Result 3 Result 4 Result 5 Result 6 x

Read Only Context Help Snippets

Object Info Session

3. Filtering with WHERE

MySQL Workbench

2520090085_Sujt x 2520090085_Vakkalanka_Sai...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- sys

Query 1

```

1 • SELECT * FROM student_marks1 WHERE marks > 85;
2 • SELECT * FROM student_marks1 WHERE marks >= 90;
3 • SELECT * FROM student_marks1 WHERE marks < 80;
4 • SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;
5 • SELECT * FROM student_marks1 WHERE name LIKE 'P%';
6 • SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');
7 • SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
8

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Filter Rows: Edit: Export/Import: Wrap Cell Contents

rol_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	15:36:13	SELECT * FROM student_marks1 WHERE marks > 85 LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
12	15:36:13	SELECT * FROM student_marks1 WHERE marks >= 90 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
13	15:36:13	SELECT * FROM student_marks1 WHERE marks < 80 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
14	15:36:13	SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90 LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
15	15:36:13	SELECT * FROM student_marks1 WHERE name LIKE 'P%' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
16	15:36:13	SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay') LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
17	15:36:13	SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%') LIMIT 0, 10...	3 row(s) returned	0.000 sec / 0.000 sec

student_marks1 7 student_marks1 8 student_marks1 9 student_marks1 10 student_marks1 11 student_marks1 12 student_marks1 13 Apply Context Help Snippets

4. Updating Queries

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Schemas list includes bookflow_db, practical_week4, practical_week_1, practical_week_2, and sys.
- Query Editor:** Contains the SQL statement:

```
1 UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
```
- SQL Additions:** A message box stating: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."
- Execution Results:** A status bar at the bottom indicates: "18 15:36:30 UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.016 sec".

5. Deleting Queries

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Schemas list includes bookflow_db, practical_week4, practical_week_1, practical_week_2, and sys.
- Query Editor:** Contains the SQL statement:

```
1 UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
```
- SQL Additions:** A message box stating: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."
- Output Window:** Shows the execution results with the following table:

#	Time	Action	Message	Duration / Fetch
1	15:37:29	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec

6. Sorting

2520090085_Suj x 2520090085_Vakkalanka_Sai...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week1
- practical_week2
- sys

Query 1

```

1 • SELECT * FROM student_marks1 ORDER BY marks ASC;
2 • SELECT * FROM student_marks1 ORDER BY marks DESC;
3 • SELECT * FROM student_marks1 ORDER BY name ASC;
4 • SELECT * FROM student_marks1 ORDER BY name DESC;
5

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

rol_no	name	subject	marks
5	Vijay	Math	80.25
2	Sita	Math	92.75
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00

Administration Schemas

Information

No object selected

student_marks1 14 student_marks1 15 student_marks1 16 student_marks1 17 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:37:46	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
2	15:37:46	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
3	15:37:46	SELECT * FROM student_marks1 ORDER BY name ASC LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
4	15:37:46	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

7. Group By

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week1
- practical_week2
- sys

Query 1

```

1 • SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
2 • SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
3 • SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

subject	num_students
Math	5

Administration Schemas

Information

No object selected

Result 18 Result 19 Result 20 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:38:04	SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
2	15:38:04	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
3	15:38:04	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

8. Having

2520090085_Suit x2520090085_Vakkalanka_Sai... x

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

SCHEMAS

Filter objects

bookflow_db

practical_week4

practical_week_1

practical_week_2

sys

Query 1

Limit to 1000 rows

1

2

3

SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject HAVING AVG(marks) > 90;

SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject HAVING COUNT(*) > 1;

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Filter Rows:

Export:

Wrap Cell Contents

subject

num_students

Math

5

Administration

Schemas

Information

No object selected

Result 23

Result 24

Read Only

Context Help

Snippets

Output

Action Output

#

Time

Action

Message

Duration / Fetch

1

15:38:29

SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject HAVING AVG(marks) > 90;

0 row(s) returned

0.000 sec / 0.000 sec

2

15:38:29

SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject HAVING COUNT(*) > 1;

1 row(s) returned

0.000 sec / 0.000 sec